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HiMCM

Summary Sheet

Abstract

Our paper models the viability of sports for their inclusion in the Olympics, using **logistic regression** to optimize the weights in our algorithm. To consider the problem, we started with the seven criteria of popularity and accessibility, gender equity, sustainability, inclusivity, relevance and innovation, safety and fair play, and, most importantly, physical exertion and entertainment. These are further broken down into 14 different metrics, which the model accepts as input. The model returns a rating signaling the applicability of the sport for the Olympics. After metrics were **normalized**, missing data was imputed by **k-nearest neighbor imputer**, and positive and negative outputs of training data were labeled. We used logistic regression to derive the metrics' weights; our model reaches an accuracy of **92.6%** in leave-one-out **cross-validation**, which shows the model's strong ability to generalize. Our model also has a **low sensitivity**, with a 10% change in raw metric data resulting in only a maximum of **6.6%** output fluctuation, thus showing the robustness of our model.

By further examining non-Olympics popular sports, our model identifies **Dance Sports, Muay Thai, and Inline Skating** as the most viable sports for the 2032 Olympics and can identify more viable sports in the future as their data change. Our model further suggests that **Karate, Boxing, and Breakdancing**, though removed from recent Olympics events, are highly fitting of the given criteria and should be reintroduced into the Olympics.

Key Words: logistic regression, normalization, k-nearest neighbor imputer, cross-validation, low sensitivity

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INTRODUCTION

Problem Summary

“Citius, Altius, Fortius”. Such is the motto of the Olympic Games, which is written in Latin yet is a global testament to what the international competition represents. As athletes strive to be higher, faster, and stronger, they continuously push the boundaries of man, challenge the extremes, and, in the process, step towards unity among diverse nations and cultures.

As such, it becomes imperative for the Olympics’ SDEs—Sports, Disciplines, or Events—to fulfill a system of criteria the IOC (the International Olympics Committee) has set. This ensures that each sport includes diverse people and acts as a medium for athletes to compete fairly. Hence, in this problem, our team is assigned the important task of building a mathematical model according to the criteria that can assist the IOC in evaluating which SDEs should be added (or potentially removed) from the 2032 Summer Olympic Games and future events.

To do so, our team will carry out a series of steps to ensure our final model is accurate and realistic. Namely, we will identify the various factors that should be considered when addressing the IOC criteria and build a model based on these factors. The model will help the IOC evaluate which SDEs align best with the Olympic criteria.

With this model, our team will test it on at least 3 SDEs that have either been recently added or removed from the Olympic Games, identify 3 SDEs that can be added to the 2032 Olympics, and perform a sensitivity analysis to observe our model’s robustness. With those aspects in mind, we will draft a letter to the IOC to summarize our findings and explain our recommendations for SDEs to either add or remove for the 2032 Brisbane Olympics.

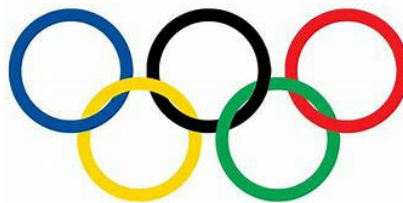


Figure 1: Olympic Ring

In the long run, our team recognizes that it is impossible to determine whether or not a sport would be permanently excluded from the Olympics, so this model mainly aims to identify the most appropriate sports to be added to the most imminent Olympics events.

Assumptions and Justifications

As our model is a simplification of the real-life situation, we will naturally have to make certain assumptions in favor of quantifying data to validate the model.

Table 1: Assumptions and Justifications

#	Assumptions	Justifications
1.	This paper will only consider sports in general categories, e.g., different forms of poker are all seen as “poker”.	It is difficult to divide the followers of one sport into different disciplines.
2.	This paper will only acknowledge the official federation of the sport, e.g., for basketball, our team will focus on studying the FIBA despite the NBA being a more prominent basketball platform.	The IOC only considers the official federation in their corporation with the sport.
3.	The IOC considers the importance of each factor the same for each sport e.g., boxing is excluded from the 2028 Olympics mainly due to issues of gender equity, but this metric tends to be less influential in consideration of other sports.	It is impossible to justify potential biases or weight discrepancies that may exist between different sports.

Variables

Table 2 shows the variables used in this paper.

Table 2: Variables

Variable	Symbol	Definitions
YouTube followers	f_y	the number of YouTube followers on the sport’s federation account
Instagram followers	f_i	the number of Instagram followers on the sport’s federation account
Twitter followers	f_t	the number of Twitter followers on the sport’s federation account
Member nations	m	the number of the sport’s federation member nations
Continental span	c	the continents that the member nation covers
VR incorporation	i	the identifier of VR incorporation of a sport
Safety	s_a	the safety score of a sport
Sustainability	s_u	the sustainability score of a sport
Anti-doping samples	a	the number of anti-doping samples of a sport done by WADA
Male-female ratio	r	the ratio between male and female of each sport
YouTube most views	v	the most views of a video on a competition of each sport
Physical exertion	p	the identifier of the need for physical movement of each sport
Entertainment	e	the score of the entertainment value of a sport
Minimum equipment cost	$c_{\min}$	the minimum cost of playing the sport
Sport vector	$\mathbf{x}$	a vector summarizing all of the factors of each sport

FACTOR EXPLANATION

Since there is no authentic dataset that directly provides all quantified information regarding the seven criteria of popularity and accessibility, gender equity, sustainability, inclusivity, relevance and innovation, safety and fair play, and physical exertion and entertainment (this final criterion is included since it also has a crucial effect on the viability of a sport for the Olympics), our team needs to set up and search for certain factors (metrics) that are both possible to find for all sports and adequately reflect the said criteria. In the end, we carefully selected 14 metrics that meet these objectives.

Popularity and Accessibility

For a sport to be viable for the Olympics, it must be well-known and widely played enough in the first place; thus, popularity and accessibility are rigid requirements for all sports for the model. To quantify the criterion of popularity and accessibility, we chose to incorporate two YouTube statistics as a summary of each sport's popularity on the internet, as other ways to measure this criterion, such as event attendance, costs, and venues available for practicing the sport, vary too much across different countries and require too much effort and time to find. We have also left statistics for other social media, such as Twitter and Instagram, to measure the criterion of relevance and innovation, as YouTube's average user age is around 35. In contrast, the two aforementioned social media have an average user age of 31, meaning that they are more representative of the younger audience's interest in the sport. Moreover, we added the minimum equipment cost metric to quantify the accessibility of each sport.

- **YouTube Followers**

One way to measure a sport's popularity is to see the number of subscribers to its official federation's YouTube account. It should be acknowledged that this factor does contain a degree of discrepancy, as the official federation account may not be the account with the most subscribers regarding the sport. However, it is the most objective way to measure popularity as it provides a balanced standard upon which we can build later calculations.

This is a deterministic, quantitative variable.

- **YouTube Most Views**

Another metric to measure the aggregate popularity of a sport among both skilled and casual players would be the highest number of views of the most-viewed YouTube video for each sport (it cannot be a YouTube Short and must feature professional play of the sport to ensure that the video is strictly representative of public interest in the sport specifically). This quantity can be easily filtered out and allows for another perspective regarding each sport's popularity on the international scale.

This is a deterministic, quantitative variable.

- **Minimum Equipment Cost**

The most direct way to demonstrate the accessibility of the sport is to show its monetary costs for a player. This metric uses an integer number to show the cost of the cheapest suitable equipment for a player to use for a sport in U.S. dollars.

This is a deterministic, quantitative constant.

Gender Equity

- **Male Female Ratio**

To address the criterion of gender equity, this factor needs to reflect the differences in males' and females' opportunities to participate. Instead of understanding how the opportunities differ between the two sexes, analyzing their resulting outcomes—the ratio between the male and female players—is a more valid way to showcase the male-female ratio. A ratio closer to 0.5 represents equal male and female opportunities.

This is a deterministic, quantitative variable.

Sustainability

- **Sustainability Score**

Sustainability, defined by the IOC, should promote environmental and social responsibility.

In its report, the IOC provides its sub-factors for analyzing sustainability as a means of environmental responsibility: infrastructure and natural sites, sourcing and resource management, mobility, workforce, and climate.

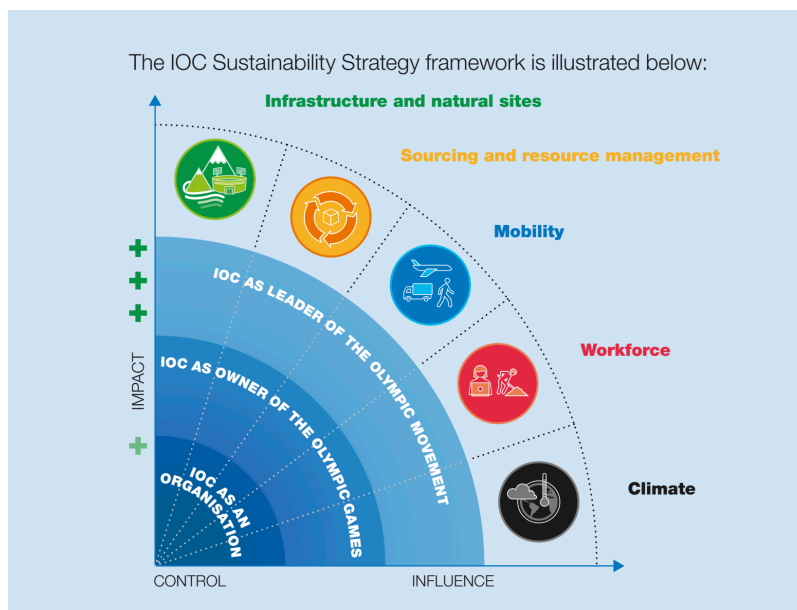


Figure 2: Sustainability sub-factors (stillmed.olympic.org)

In the listed subfactors, our team combined sourcing and resource management with mobility to form the whole subfactor of resource allocation. For the workforce, this subfactor is separated from the qualitative analysis of the other factors. Also, the factor infrastructure and natural sites will be called infrastructure for short.

The IOC did not provide specific subfactors to analyze sustainability through social responsibility. Our team generally decided to split social responsibility into its social impact and physical health. Social impact is designed to be sports' effect on promoting good behaviors, while physical health is designed to be sports' effect on promoting good health.

In total, five subfactors affect the sustainability factor: infrastructure (and natural sites), resource allocation, climate, social impact, and physical health.

Table 3: Sustainability scoring guideline

Factor	Definitions	Points
infrastructure	The amount of land usage	2
resource allocation	The amount of resources needed and carried	2
climate	The impact on the climate	2
social impact	The positive impact on behaviors	2
physical health	The positive impact on health	2

Our team uses a scoring guideline to measure this data to score with sports within the five sub-factors. Adding them up and dividing them by ten forms the score that each sport will receive in the sustainability factor portion.

This is a deterministic, qualitative variable.

Inclusivity

The IOC defines inclusivity as the criteria for “representing diverse cultures and promoting global participation.” Hence, to analyze this aspect more thoroughly, the following factors will be considered: the number of member nations and the continental span.

• Number of Member Nations

First, the number of member nations refers to the number of nations that are members of the respective sports’ international federation. As such, this factor will provide a fundamental understanding of each sport’s degree of global participation. Since the Olympic Games pose as an international stage for competition, various nations must be included for a diverse pool of athletes.

Moreover, it is important to note that the IOC recognizes the international sports federations that administer the sport in question as one at the world level. Taking the number of member nations into account will also filter out sports that do not meet the standard of international inclusivity.

This is a deterministic, quantitative variable.

• Continental Span

It has already been pre-stated that at least 75 countries across four continents must practice the sport to be acknowledged as a potential Olympic sport. The former condition– at least 75 countries– has already been considered from the previous factor. The continental span will then be measured quantitatively to determine the number of continents that practice the sport. As a quantitative, deterministic constant, this factor will ensure that the criterion of inclusivity is thoroughly analyzed.

This is a deterministic, quantitative variable.

Relevance and Innovation

Unlike YouTube users, who are generally 35 years old, Twitter and Instagram users are generally younger (around 31 years old) and can represent the young audience appeals the IOC aims for.

- **Twitter Followers**

This measure is a deterministic, quantitative variable represented by the number of followers on the official federation. Similar to the metric of YouTube followers, this metric measures the sport's popularity among skilled players, but the younger age of the followers gives a different perspective on the sport's relevance and innovation.

This is a deterministic, quantitative variable.

- **Instagram Followers**

This metric of the followers of official sports federations has the same logic as the metric of Twitter followers. It helps to complete some information missing from "Twitter followers," as some federations do not have a Twitter account.

This is a deterministic, quantitative variable.

- **VR Incorporation**

The IOC's criterion of relevance and innovation requires incorporating innovations such as physical and virtual sports while satisfying Olympic traditions.

The only measure to measure VR Incorporation has only one possible factor: the progress of physical and virtual sports, which is a complex measure. To simplify the physical, virtual sports factor, our team chose to round the progress of physical, virtual sports into a boolean factor on whether there is a basic VR game for this sport that respects Olympic traditions.

This is a deterministic, qualitative variable.

Safety and Fair Play

- **Safety**

The IOC defines the safety portion of Safety and Fair Play as maintaining high standards for athlete protection.

An obvious measure for this criterion is injury rates. However, it is not enough to measure safety because the injury level of each injury is different. An injury to the head is different from a strain. So, to measure the safety of sports, the injury level must also be considered.

Since data for each sport's injury level is difficult to find, our team implemented a strict scoring guideline based on injury level.

Table 4: Safety Sub Factors (IOC Reports)

Factor	Definitions	Points
acute injuries	Minor injuries such as sprains	2
head injuries	Severe injuries such as concussion	3
joint injuries	Middle injuries such as breaking bone	2

Our team uses a scoring guideline to measure this data to score with sports within the three sub-factors. We are adding them up and dividing them into seven forms, with each sport receiving a score in the sustainability factor portion.

• Anti-Doping Samples

The IOC defines the safety portion of safety and fair play as maintaining high standards for anti-doping.

The IOC forms a relationship with the World Anti-Doping Agency (WADA) to govern anti-doping samples on sports.

By researching the anti-doping samples on the World Anti-Doping Agency, it is possible to get most sports' anti-doping statistics. The number of anti-doping samples reflects the sport's anti-doping policies, showing that the federation is working hard on diminishing doping issues.

This is a deterministic, qualitative variable.

Physical Promotion and Entertainment

Even though this criterion was not listed among the six criteria the IOC uses for identifying appropriate Olympic sports, our team strongly believes this criterion is crucial to identifying an appropriate sport. In fact, it might be the most important, as most sports were first developed without aiming for other equity, such as gender equity or safety, but for spectacle and physical exercise. Therefore, the early Olympic games, such as equestrian and aquatics, were mainly picked for these two qualities, and they remain one of the most important qualities for the participants and audience of the Olympics.

Moreover, in the data we use in the model's training, the sports labeled as appropriate for the Olympics are only so since they are already in the event. This means that the aspects of spectacle and physical exercise are already inherent in the model's training set, and thus, it is absolutely necessary to include metrics for such aspects to make the model's predictions for appropriate sports more adherent to the previous patterns of the Olympic games.

• Physical Exertion

This factor is a qualitative factor used to distinguish other sports from mental sports and machine sports, such as chess, poker, sports car racing, water motorsports, etc. As the concepts of sport and Olympicism are inherently and deeply associated with the element of physical exercise and challenges, this metric makes a crucial gatekeeper for the qualification of any Olympic sport.

This is a deterministic, qualitative constant.

- **Entertainment**

This factor is a qualitative factor used to exclude sports that are not entertaining as other sports, therefore not in the Olympics, such as chess, cornhole, and go. Since the Olympics is designated as an event to be viewed by billions of people worldwide, it must ensure that its included sports are aesthetically pleasing or entertaining.

This is a deterministic, qualitative constant.

MODEL

Data Processing

Normalization

Normalization is the process of adjusting different values into one scale. In this model, our team modified all the values to fit in the range $[0, 1]$ using different normalizers. Our team used four normalizers:

1. Limited Domain Normalizer
2. Infinite Domain Normalizer
3. Conditional Infinite Domain Normalizer
4. Distance Normalizer

Limited Domain Normalizer

This normalizer, acting on the domain $[0, r]$, increases the upper bound r such that the range of the normalizer becomes $[0, 1]$ after division.

$$\text{LDN}(x, r) := \left\{ \frac{x}{r} \mid 0 \leq x \leq r \right\} \quad (1)$$

This normalizer is used to normalize sets with a specific domain, and it has the property that the normalized value increases alongside the raw factor quantity.

This normalizer is used to normalize the factors:

1. VR incorporation (i)
2. Sustainability (s_u)
3. Physical exertion (p)
4. Entertainment (e)

Infinite Domain Normalizer

This normalizer, acting on the domain $[0, \infty)$ uses a function to return results in the range of $[0, 1]$.

$$\text{IDN}(x) := \left\{ 1 - \frac{1}{1 + \ln(1 + x)} \mid x \geq 0 \right\} \quad (2)$$

Our team has considered two other infinite domain normalizers: tanh function($\frac{e^x - e^{-x}}{e^x + e^{-x}} \mid x \geq 0$), arctan function($\frac{2}{\pi} \arctan(x) \mid x \geq 0$). Comparing the three functions, the inverse function we are using now increases the slowest. To make the inverse function flatter, our team replaced the x in the inverse function with $\ln(x + 1)$.

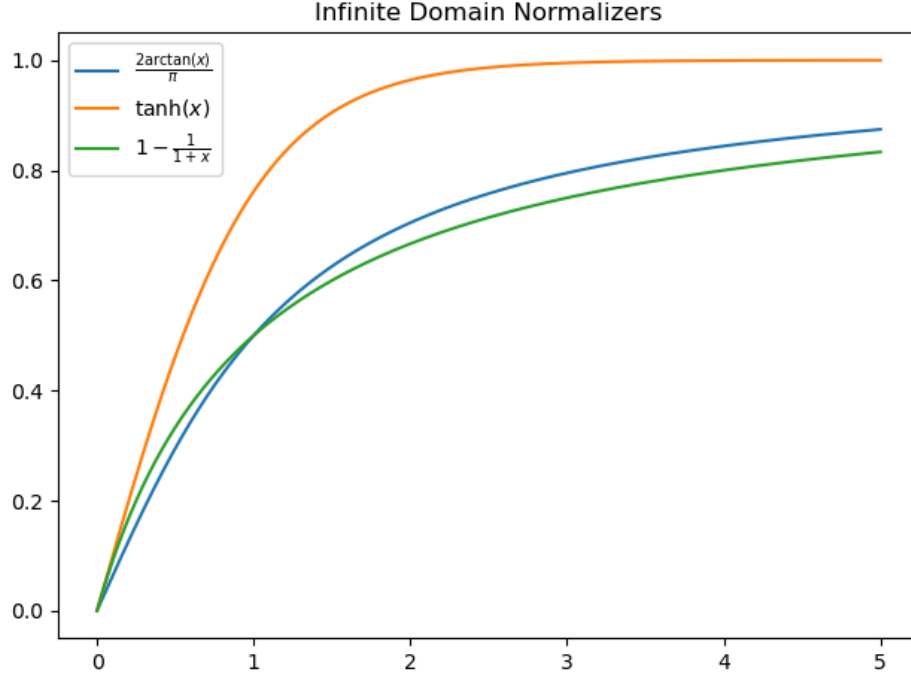


Figure 3: The three infinite domain normalizers

The purpose of this normalizer is to normalize sets without a specific domain, with the property that the normalized value increases alongside the raw factor quantity.

This normalizer is used to normalize the factors

1. YouTube followers (f_y)
2. Instagram followers (f_i)
3. Twitter followers (f_t)
4. YouTube most views (v)
5. Anti-doping samples (a)
6. Minimum equipment cost ($c_{\min}$)

Conditional Infinite Domain Normalizer

This normalizer, acting on the domain $[0, \infty)$, sets values lower than a threshold value to zero and outputs in the range of $[0, 1]$.

$$\text{CIDN}(x, b) := \begin{cases} 0 & 0 \leq x < b \\ 1 - \frac{1}{1 + \ln(1+x-b)} & x \geq b \end{cases} \quad (3)$$

The purpose of the normalizer is to make sure that numbers lower than the threshold will not be considered in the algorithm, which magnifies differences between certain sports on special factors e.g. sports with only 3 or 4 continents influenced are, in truth, not inclusive, but is numerically close to 5 and 6. Otherwise, it is the same as Infinite Domain Normalizer.

This normalizer is used to normalize the factors

1. Continental Span (c)
2. Member Nations (m)

Distance Normalizer

This normalizer, acting on the domain $[0, 1]$, prioritizes the elements at the middle, and results in the range of the normalizer to be $[0, 1]$.

$$\text{DN}(x) := \{1 - 4(x - 0.5)^2 \mid 0 \leq x \leq 1\} \quad (4)$$

The purpose of the normalizer is to make the numbers that are close to 0.5 to be the largest, mainly targeted on values that focus on equality.

This normalizer is used to normalize the factors

1. Safety (s_a)
2. Male Female Ratio (r)

For Male female ratio, FRAC is applied before the application of $\text{DN}(x)$ to make the domain between $[0, 1]$:

$$\text{if } r = a : b, \text{ then } \text{FRAC}(r) = \frac{a}{a + b} \quad (5)$$

The variable safety using the distance normalizer is that the Olympic games are never considered completely safe mental sports, and too-safe sports, such as frisbee, often lack physical exertion. Thus, the safest and most dangerous sports are both less likely to be considered by the IOC.

Data Imputation

While collecting data, we inevitably encountered the absence of data for a specific sport. Initially, we decided to delete every sports sample with missing data, but then we realized that it would be quite a waste if a sample had only 1 factor's data missing out of the 13 factors. We decided to impute the missing data while trying to preserve the distribution of the dataset. To do this, we used the k-nearest-neighbor method to impute the data.

Each sport sample is essentially a vector of values, $\mathbf{x}$. The L_2 distance or euclidean distance of two samples $\mathbf{x}$ and $\mathbf{y}$ is defined as $\sqrt{\sum_{i=1}^m (x_i - y_i)^2}$.

The k-nearest neighbor method finds the k samples with the least Euclidean distance with the sample with the missing factor's data and imputes the missing factor by taking the average of the k sample's own factors. (It must be ensured that the k samples all have the missing factor's data). Our model uses $k = 5$ for data imputation.

Normalization Results

The data can be used in our model by normalizing each factor with its specific normalization function. The normalization process is summarized below, where x_{sport} is a vector containing all the elements.

$$\begin{aligned} x_{\text{sport}} = & \\ & (\text{IDN}(f_y), \text{IDN}(f_i), \text{IDN}(f_t), \text{IDN}(c_{\min}), \text{CIDN}(m, 75), \text{CIDN}(c, 4), \text{LDN}(i, 1), \text{DN}(\text{LDN}(s_a, 7)), \\ & \text{LDN}(s_u, 10), \text{IDN}(a), \text{DN}(\text{FRAC}(r)), \text{IDN}(v), \text{LDN}(p, 1), \text{LDN}(e, 2)) \end{aligned} \quad (6)$$

In the following diagrams, the normalization outcomes of r , s_u , and i indicate the data distribution and imply the metrics' effectiveness for judging the Olympic viability of sports.

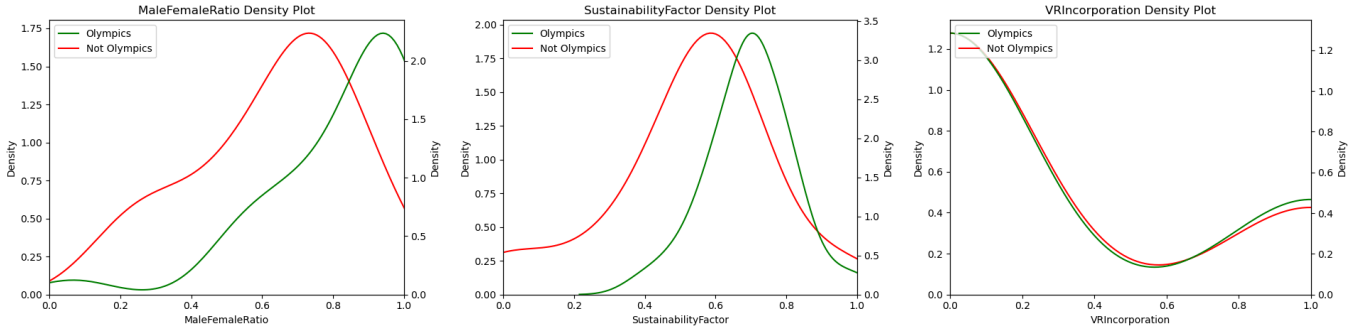


Figure 4: Density Plots of Male and Female Ratio, Sustainability, and VR Incorporation

The green line represents the density of the Olympic-viable sports, while the red line represents the density of the sports unviable for the Olympics. If most of the area under a curve leans to the right side, then the sports represented by the curve would have generally more desirable ratings in the metric. For instance, there is an apparent difference between viable and unviable sports in male-female ratio: Olympic-viable sports have generally more desirable (more fit for gender equity) male-female ratios than non-Olympic sports. The sustainability factor curves are similar to that of the male-female ratio. The difference in the ratings implies that both factors play a significant role in identifying Olympic and non-Olympic sports. Meanwhile, there is almost no difference in VR incorporation, indicating that VR incorporation is an insignificant factor to consider.

Algorithm

Each sample in the normalized dataset can be represented by vector $x = (x_1; x_2; \dots; x_d)$ with length d , where x_i is the value of x on the i th factor. We determined that the evaluation function $f(x)$, which outputs the rating that judges if a sport should be included in the Olympics, should satisfy the following requirements:

1. $f(x) \in [0, 1]$
2. $f(x)$ should be closer to 1 if x represents a sample that makes an excellent Olympic sport, and closer to 0 if x represents a sample that does not.

3. The evaluation process should have good comprehensibility: its values should be explainable.

To satisfy requirement 3, we decided to use a linear combination of weights and factors to express our model:

$$g(\mathbf{x}) = w_1x_1 + w_2x_2 + \dots + w_dx_d + b = \mathbf{w}^T \mathbf{x} + b \quad (7)$$

However, the linear model $g(\mathbf{x})$ does not fit in the range $[0, 1]$ mentioned in requirement 1. To address this requirement, we need to choose a function that normalizes the $g(\mathbf{x})$ output to the range $[0, 1]$. There are many choices, such as the unit-step function $h(z) = \begin{cases} 0, & z < 0 \\ 0.5, & z = 0 \\ 1, & z > 0 \end{cases}$, the inverse tangent function $h(z) = \frac{1}{\pi} \arctan(x) + \frac{1}{2}$, or the logistic sigmoid function:

$$h(z) = \frac{1}{1 + e^{-z}} \quad (8)$$

Ultimately, we chose the logistic sigmoid (expit) function as the normalization function due to its differentiability and simplicity. Thus, the evaluation function becomes:

$$f(\mathbf{x}) = h(g(\mathbf{x})) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (9)$$

To solve for $\mathbf{w}$ and b , we need example sports that serve either as positives (those that are already included in the Olympics) or negatives (those that are publicly denied, such as chess and motorsports, or those that have been inconsistent or are insufficient). We label sports that we are certain are Olympics-qualified as '1's and those not as '0's. Examples of '1's would be the sports consistently included in the Olympics, such as athletics and basketball, and examples of '0's would be those that are openly rejected by the IOC, such as chess and sports-car racing, or those that completely fail to fulfill certain criteria, such as poker (no physical exertion) and Wushu (not inclusive). These labelled samples together form the model's training dataset $D_{\text{train}} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_m, y_m)\}$ where $\mathbf{x}_i$ is the i th sample and y_i is the label of the corresponding sample.

We then used logistic regression to optimize for $\mathbf{w}$ and b . $y = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}}$ can be rewritten as:

$$\ln \frac{y}{1 - y} = \mathbf{w}^T \mathbf{x} + b \quad (10)$$

If we treat y as the probability of sample $\mathbf{x}$ having a label of 1, the above equation can be transformed into:

$$\ln \frac{p(y = 1|\mathbf{x})}{p(y = 0|\mathbf{x})} = \mathbf{w}^T \mathbf{x} + b \quad (11)$$

We can then use the maximum likelihood method to estimate $\mathbf{w}$ and b . Let $\beta = (\mathbf{w}; b)$ and $\hat{\mathbf{x}} = (\mathbf{x}; 1)$, then $\mathbf{w}^T \mathbf{x} + b = \beta^T \hat{\mathbf{x}}$. The target function to optimize is:

$$\ell(\beta) = \sum_{i=1}^m \left(-y_i \beta^T \hat{x}_i + \ln(1 + e^{\beta^T \hat{x}_i}) \right) \quad (12)$$

The optimal parameters are:

$$\beta^* = \arg \min_{\beta} \ell(\beta) \quad (13)$$

We can use the Newton method to optimize for β :

$$\beta' = \beta - \left(\frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T} \right)^{-1} \frac{\partial \ell(\beta)}{\partial \beta} \quad (14)$$

Where

$$\begin{aligned} \frac{\partial \ell(\beta)}{\partial \beta} &= - \sum_{i=1}^m \hat{x}_i \left(y_i - \frac{e^{\beta^T \hat{x}_i}}{1 + e^{\beta^T \hat{x}_i}} \right) \\ \frac{\partial^2 \ell(\beta)}{\partial \beta \partial \beta^T} &= \sum_{i=1}^m \hat{x}_i \hat{x}_i^T \frac{e^{\beta^T \hat{x}_i}}{(1 + e^{\beta^T \hat{x}_i})^2} \end{aligned} \quad (15)$$

To prevent the model from becoming too sensitive, we also introduced L_2 regularization, adding the term $\frac{\|w\|_2^2}{2C}$ to $\ell(\beta)$, where C is the regularization strength.

We used the leave-one-out cross validation method to test our model's ability to generalize to divide the dataset into training and testing sets. The leave-one-out method takes one sample as the testing set and the rest as a training set and trains an individual model on every single train-test set combination. The accuracy of the model is then computed from the model's score on each train-test set combination. The leave-one-out is often seen as relatively accurate at measuring the generalizability of a model. Our model achieved an accuracy of 92.6%. This accuracy is acceptable, so the model is now trained on the entirety of the dataset for optimal performance. The weights and bias are as follows:

Table 5: Weights and Bias

Factor	Weight
YouTube followers	0.162590
Instagram followers	0.537694
Twitter followers	1.298255
Minimum equipment cost	-0.112346
Member nations	1.551941
Continental span	0.092116
VR incorporation	0.316370
Safety	0.838445
Sustainability	0.901123
Anti-doping samples	0.732160
Male-female ratio	0.507280
YouTube most views	0.006624
Physical exertion	1.655687
Entertainment	0.170130
Bias	-5.905663

Physical exertion is the most important criterion for deciding which sports are Olympic in our model. This makes sense since all sports in the Olympic Games are physical sports.

The next factor that has the largest weight is member nations, which measures the inclusivity of the sport. This parallels the countries participating in the sport, which the IOC requires strictly. It is evident that the IOC wants sports with a lot of countries participating.

The third important factor is Twitter followers. This can be justified by the recent addition of skateboarding and surfing, which are all sports with younger audiences. Twitter followers are the factor used to measure the sport's popularity among young people.

One crucial factor to note is the minimum equipment cost, which holds a negative weight. This is because a lower cost signifies higher accessibility and is more accessible, so the weight must be negative to ensure that this conclusion stands true.

Overall, the factors' weights are reasonable.

To evaluate any given sport sample with the evaluation function, normalize the sample using the normalization functions defined in the last section, and apply the function below.

$$\mathbf{w}^T = (0.163, 0.538, 1.298, -0.112, 1.552, 0.092, 0.316, 0.838, 0.901, 0.732, 0.507, 0.007, 1.656, 0.170) \quad (16)$$

$$b = -5.905663 \quad (17)$$

$$f(x) = \frac{1}{1 + e^{-(w^T x + b)}} \quad (18)$$

If $f(x) \geq 0.5$, then the model predicts that the sport will be in the Olympics.

TESTING THE MODEL

To test our model's accuracy, we selected some sports with known viability in the Olympics as the testing set.

Recently Added / Removed Sports

For recent changes, our team chose to test skateboarding, cricket, flag football, and boxing.

1. **Skateboarding** is a sport with gradually rising popularity and relatively high risk. In 2020, the IOC approved it for the Olympics. Skateboarding, along with breakdancing and surfing, represents the culture of the younger generations.
2. **Cricket** is a popular sport, but its popularity is centered in a few regions. It only has around 100 member nations, with only a few being highly active. Cricket represents a sport that is very popular in small regions, yet still managed to get into the 2028 Olympics due to its other criteria.
3. **Flag Football** is a version of American Football known for high injury rates (but not as high as American Football). Yet, due to its rise in global popularity, flag football still managed to make it into the 2028 Olympics.
4. **Boxing** has been a sport in the Olympics since 1988, but it was partially removed from the 2028 Olympics because of the notorious scandals that happened in the 2024 Olympics. Our model did not consider this scandal.

The four sports are examples of significant advantages and disadvantages in different categories, making it difficult to classify them correctly. Thus, this tests the robustness of our model.

Consistent Sports

Our team tests archery, basketball, and weightlifting for consistent sports.

1. **Archery** is a relatively less popular sport than other sports. This test sample can check whether the model has a well-designed acceptance of less popular sports.
2. **Basketball** is a popular sport that must be in the Olympics. This sport can represent all of the common, popular sports that have always been in the Olympics.
3. **Weightlifting** is a sport that does not rely on movement. This kind also includes shooting, gymnastics, and breaking (which was removed from the 2028 Olympics).

These three sports represent different categories of sports that have consistently been in the Olympics, and testing them reflects the consistency of our model.

Rejected Sports

Some sports have been considered part of the Olympic sports but ultimately failed. One example is chess.

1. **Chess** is a popular, recognized sport with millions of fans, yet it has never entered the Olympics. Chess represents a mental sport that has never entered the Olympics.
2. **Water Motorsports** is a recognized sport with substantial viewership but expensive equipment. It was featured once in the 1908 Olympics. It is representative of motorsports that are more focused on machinery and vehicles and often less on physical exercise, and it lacks accessibility and sustainability.

The rejected sports all feature extreme failures in specific criteria, such as sustainability, physical exertion, or other factors.

Testing Results

The model’s rating results are shown in the following table.

Table 6: Testing Results

Sport	Value	Correct
Skateboarding	0.766057	1
Cricket	0.768819	1
Flag Football	0.646510	1
Boxing	0.757936	0
Archery	0.792439	1
Basketball	0.823031	1
Weightlifting	0.845631	1
Chess	0.165358	1
Water Motorsports	0.094497	1

The testing results match the expected relative value difference between consistent sports and recently added/removed sports. The consistent sports have an average rating of 0.820, while the recently added sports have an average rating of 0.735. This shows that our model gives reasonable ratings, considering that sports consistently included in the Olympics should naturally secure firmer positions in the Olympics.

The only mistake in the testing sample is boxing. Although it has a relatively low rating compared with other sports that stayed in the Olympics, it is still viable in the Olympics due to its evaluation being greater than 0.5. The reason that the Olympics did not include boxing is the gender scandal that happened in the 2024 Olympic Games, leading to severe governance issues that have yet to be solved. This is a problematic factor that our model did not include.

The main reason chess and water motorsports are not in the Olympics is that the IOC wants to maintain the prominence of its core value of Olympism (exercise and performance) in its events.

Overall, our model is highly accurate in predictions of sports in the Olympics.

MODEL PREDICTION RESULTS

Candidates for Olympics Introduction

Our team selected some possible sports that have popularity and which have the potential to be in future Olympic games, which are Bowling, Billiard Sports, Muay Thai, Sumo, American Football, Darts, Ballet, Frisbee, Dodgeball, Parkour, Yoyo, Inline Skating, Cornhole, and Dance Sports.

After our model is run on these 14 sports, some sports are recognized by our model as viable for the Olympics game.

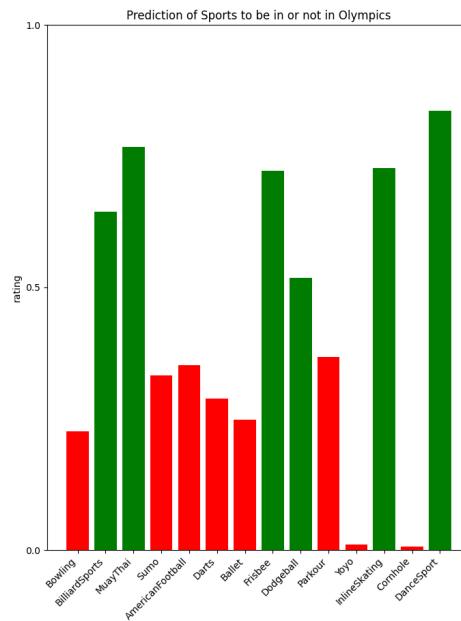


Figure 5: Prediction of Sports to be in or not in Olympics

Six sports have been predicted to have the ability to be admitted into the Olympic games.

1. Dance Sport ($f(x_{\text{Dance Sport}}) = 0.834711$)
2. Muay Thai ($f(x_{\text{Muay Thai}}) = 0.764858$)
3. Inline Skating ($f(x_{\text{Inline Skating}}) = 0.725549$)
4. Frisbee ($f(x_{\text{Frisbee}}) = 0.723292$)
5. Billiard Sports ($f(x_{\text{Billiard Sports}}) = 0.642635$)
6. Dodgeball ($f(x_{\text{Dodgeball}}) = 0.518257$)

Future Olympics Inclusion

• For 2032

Our team proposes three candidates for the 2032 Olympics: Dance Sports, Muay Thai, and Inline Skating.

1. **Dance Sport** is a recognized sport by the IOC that has not entered the Olympics. Although similar to gymnastics and breaking, dance sport scores higher than breaking and close to gymnastics, inferring that it meets the IOC's criteria.
2. **Muay Thai**, commonly called Thai Boxing, is a sport similar to boxing and karate. Its federation, the International Federation of Muay Thai Associations, has 128 member nations, which shows that it is a regional sport like American Football.
3. **Inline Skating** is a sport which holds more popularity among the younger population. As the IOC is focusing more on appealing to younger audiences, inline Skating, as predicted by the model, should be in the Olympics.

• For 2036 or Beyond

Frisbee, Billiard Sports, and Dodgeball can be considered for future Olympic games. Although their scores are lower than those of Dance Sports, Muay Thai, and Inline Skating, they are still predicted to be viable for the Olympics.

Meanwhile, sports removed from the Olympics, namely boxing, karate, and breaking, all have high ratings according to our model. They left the Olympics for special reasons, such as scandals and choosing to host cities. This justifies that these sports fit the six criteria provided by the IOC and contain sufficient physical exertion and entertainment, which could be reintroduced in the future. As they said, societal disputes settle down.

SENSITIVITY ANALYSIS

We conducted a sensitivity analysis regarding our model. First, we computed the average of each factor of our samples as the ‘default sport’. Then, the score of the ‘default sport’ was evaluated with the model. Then, each factor was altered by $\pm 10\%$, and the resulting changes in the model are less than 6.6%, which proves our model’s robustness and stability.

Table 7: Sensitivity Analysis Results

Factor altered	Ratio to standard value	Model output	Percent change
/	100%	0.5152531880441339	0%
YouTube followers	90%	0.5121928601135086	-0.59%
YouTube followers	110%	0.5183123724946884	0.59%
Instagram followers	90%	0.5046025338148539	-2.07%
Instagram followers	110%	0.5258900052411963	2.06%
Twitter followers	90%	0.4918641624431917	-4.54%
Twitter followers	110%	0.5385755881664331	4.53%
Minimum equipment cost	90%	0.517356495443733	0.41%
Minimum equipment cost	110%	0.5131493401739615	-0.41%
Member nations	90%	0.49253503580115787	-4.41%
Member nations	110%	0.5379084769284349	4.40%
Continental span	90%	0.513976030771407	-0.25%
Continental span	110%	0.5165301461229806	0.25%
VR incorporation	90%	0.5131172755386274	-0.41%
VR incorporation	110%	0.5173885434811365	0.41%
Safety	90%	0.5023551981988569	-2.50%
Safety	110%	0.5281308910026575	2.50%
Sustainability	90%	0.5007973383876377	-2.81%
Sustainability	110%	0.5296835590694996	2.80%
Anti-doping samples	90%	0.5010530039548412	-2.76%
Anti-doping samples	110%	0.5294287859944231	2.75%
Male-female ratio	90%	0.5063151871534883	-1.73%
Male-female ratio	110%	0.5241814421381129	1.73%
YouTube most views	90%	0.5150973788366898	-0.03%
YouTube most views	110%	0.5154089942865216	0.03%
Physical exertion	90%	0.48170487385570854	-6.51%
Physical exertion	110%	0.5486645968695815	6.48%
Entertainment	90%	0.5114624856259434	-0.74%
Entertainment	110%	0.5190421362048864	0.74%

LETTER

Dear International Olympic Committee,

Good day, and we hope this letter finds you well.

For centuries, the Olympic Games have represented man's continuous challenge of the extreme. And since your esteemed committee's founding in 1894, the Olympics have undergone revolutionary changes that define what it is today. As such, our team has constructed a mathematical model to evaluate whether a sport can be accepted as an Olympic sport. It bases itself upon the criteria the IOC has laid out according to the Olympic values and considers various factors to ensure maximal realism. Hence, through this letter, our team strives to provide you with a better understanding of our model, and we genuinely hope it can aid your esteemed committee's advances in building a more diverse and healthy athletic community.

It is important to lay some foundations first. To better target each criterion the IOC has set, our team identified multiple factors that play a significant role in defining the standards.

Regarding Popularity and Accessibility, we focused on three main factors: the number of YouTube followers for each official sports federation, the highest amount of views for each most-viewed sports video, and the minimum cost to participate in each sport. For Gender Equity, we researched the ratio between male and female professional players of each sport. The sustainability criterion was subdivided into smaller infrastructure sections, resource allocation, climate, social impact, and physical health. For Inclusivity, which emphasized the ability to include diverse cultures across nations, our team listed two factors: the number of member nations for each International Sports Federation (ISF) and the continental span of each sport. Relevance and Innovation, on the other hand, stress the appeal to younger audiences, so we analyzed the number of followers for each sport's official account on both Instagram and Twitter—as they sported a younger average audience—and each sport's tendency to incorporate VR technology. For Safety and Fair Play, we analyzed injury chances and anti-doping samples.

After conducting research, though, our team decided to include two other criteria to be considered: Level of Physical Exertion and Entertainment Value. We recognize that the Olympics not only poses as an international competition but also as the host for a large audience for every game. This inclusion ensures that the final results are accurate and as realistic as possible.

With this in mind, our team continued utilizing the data collected to construct the mathematical model. It combines all the aforementioned metrics and compares their weights against each other to produce a final value between 0 and 1—1 being the best choice and 0 being the worst. We labeled the sports with a value above 0.5 as an appropriate choice for Olympic sports.

To test the model, our team then identified four Olympic sports that have either been removed or introduced within the last three Olympic years: skateboarding, Cricket, Flag Football, and Boxing. It should be noted that those sports all feature unbalanced disadvantages and advantages, so this test also proves our model's robustness. Moreover,

after running through all four sports through the model and comparing their values with those of consistent sports, the differences match up; our model is successful.

Furthermore, since our model is designed to be simple—after training, it only features simple addition and multiplication—it is highly comprehensible. It could be easily explained to make sense of human intuition. This low level of abstraction enables our model to be used by non-experts and has broader applications in sports than other models.

Our team then identified a list of sports never admitted to the Olympic Games to run through the model; out of 14 sports, six featured values exceeding 0.5. Combined with the concept that the sport with a value closest to 1 would be the most appropriate choice, the three SDEs our team recommends to be instated as a sport in the 2032 Olympics are Dance Sport, Muay Thai, and Inline Skating. This is listed in order from the sport to be considered first for inclusion to the third.

Additionally, we found that the three sports removed from recent Olympics events, i.e., karate, boxing, and break-dancing, have high ratings and fit the core criteria and values of the Olympics. We understand these sports have been removed for appropriate and valid reasons due to societal disputes. However, we believe that they remain closely integrated with Olympic values of unity and exercise and are fit to be reintroduced in future events.

This concludes our team's model and evaluation results, and we genuinely hope it is of use to you.

Best Regards,

Team 15715

STRENGTHS AND WEAKNESSES

After evaluation, it is identified that our model has the following main strengths:

1. Our model takes into consideration multiple factors for each criterion listed by the IOC. For instance, regarding the criterion of relevance and innovation, society's attitude towards different sports and their versatility in VR incorporation are studied through extensive numbers research. By doing so, it can be generally ensured that our model includes all the vital components for each criterion, hence achieving more realistic results.
2. Our model considers the presence of additional criteria—physical exertion and entertainment—to achieve a more realistic model. The two factors are essential to why mental sports are excluded from the Olympic games.
3. Our model is flexible and able to adapt to drastic changes in data. For example, specific statistics, such as gender equity and sustainability, might change alongside technological and societal developments. However, our model could easily update the weight of each factor and maintain reasonable results, as it is simple and highly comprehensible.

Likewise, our model also contains a few weaknesses:

1. Our model cannot consider the changes in bias and opinions of people of specific sports. For example, the 2024 Olympic Boxing gender scandal led to severe controversies about the viability of boxing as an Olympic sport, and boxing was removed from the 2028 Olympics. These events are unpredictable but hold considerable weight in the IOC's decision to admit a sport into the Olympics.
2. When researching the different factors, there were some aspects for which we could not find the corresponding data. This is either because the sport leans greatly towards the unpopular scale or because the information simply did not exist. As such, it becomes inevitable for some spaces to be left blank during our initial research process, which would lead to some slight deviations in the model. However, our datasheet has minimal blanks, and our team implemented data imputation, so the deviation will not pose a significant issue in our model.
3. Due to the special nature of the sports field and the tight time limit, our team could only gather limited metrics and data regarding a limited number of popular sports (below 100). The low quantity of sports popular enough to be considered by the IOC makes our model's training dataset very limited. It prevents us from making it more aggregate and accurate when considering sports. However, this problem could be diluted as more sports become popular and data becomes more abundant. It is also important to note that our model maintains high accuracy even under such unfavorable dataset conditions.

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Figure 6: Websites uses

APPENDIX

	Category	Federation	Region/Tournament	FollowersOnYouTube	FollowersOnInstagram	FollowersOnTwitter	MemberNumbers	ContinentalSoc.	Videos/Posts	Sustainability	actor	NumberOfChampions	Medal	emissions	InfrastructureCost	Production/Services	Physicalization	Environment	Label	
Aquatics	consistent	World Aquatics	World Aquatics Swimming World Cup	376000	1450000	138900	218	6	0	0.5174285714	0.8	15945	55,744.3		\$100	2404824	1	2	1	
	consistent	World Archery	World Archery Championships	610000	330000	81200	140	0	1	0.8174285714	0.6	919	68,153.1		\$0	6181811	1	2	1	
Athletics	consistent	World Athletics	World Athletics Championships	2130000	2900000	115300	204	6	0	0.5174285714	0.8	3390	54,946.0		\$0	4662278	1	2	1	
	consistent	IFBA	IFBA Basketball World Cup	286000	210000	213	6	0	0.8174285714	0.6	178	84,827.7		\$10	1304428	1	2	1		
Badminton	consistent	BWF	BWF World Championships	1230000	2000000	43100	201	6	1	0.8174285714	0.7	1761	5098,3499		\$100	1304748	1	2	1	
	consistent	ICF	ICF Canoe Sprint World Championships	328000	90000	150	6	1	0.8174285714	0.6	4707	53,047.0		\$200	1030713	1	2	1		
Cycling	consistent	UCI	Tour de France	390000	1100000	36400	6	1	0.4285714286	0.7	2251	15,146.8		\$10	26479135	1	2	1		
	consistent	FEI	FEI World Equestrian Games	377000	561000	105400	130	6	0	0.28174286	0.6	951	37,633.0		\$400	1252971	1	1	1	
Field Hockey	consistent	FIH	World Hockey Championships	170	6	0	0.5174285714	0.8	140	0	0.5174285714	0.6	181	401,403		\$100	1044187	1	2	1
	consistent	FIH	Men's FIH Hockey World Cup	338000	357000	12600	130	6	0	0.4285714286	0.5	154	23,077.2		\$10	139866	1	2	1	
Football	consistent	IFFA	IFFA World Cup	2200000	420000	1460000	211	6	0	0.4285714286	0.7	1860	11,226.0		\$40	6746702	1	2	1	
	consistent	IFP	IFP Championship	610000	97000	32600	132	6	0	0.8174285714	0.5	167	62,128.0		\$200	7668302	1	2	1	
Gymnastics	consistent	FIG	Symposium World Championships	903000	490000	92100	161	6	0	0.4285714286	0.1	3059	3,022.0		\$0	24777505	1	1	1	
	consistent	IFP	Paris World Cup	390000	251000	64600	60	6	0	0.4285714286	0.6	460	1004,1073		\$400	1350404	1	2	1	
Judo	consistent	IFJ	World Judo Championships	514000	130000	99700	208	6	0	0.4285714286	0.8	424	79,321.0		\$0	984143	1	2	1	
	consistent	World Lacrosse	World Lacrosse Championship	3200	6147	21800	61	6	0	0.4285714286	0.6	146	82,18.0		\$100	138887	1	2	1	
Modern Pentathlon	consistent	UPM	World Modern Pentathlon Championships	7270	20300	6171	6	0	0.28174286	0.7	718	1,01.0		\$118	1570087	1	2	1		
	consistent	World Rowing	World Rowing Championships Regatta	57600	228000	40900	150	6	1	0.8174285714	0.7	4877	55,045.0		\$200	180758	1	2	1	
Rowing	consistent	World Sailing	Sailing World Championships	446000	150000	12000	140	6	0	0.4285714286	0.1	769	84,126.0		\$200	471569	1	2	1	
	consistent	ISSF	ISSF World Shooting Championships	152000	86000	32300	163	6	1	1	0.6	2011	68,032.0		\$100	1405126	1	2	1	
Shooting	consistent	ITTF	World Table Tennis Championships	51800	50700	18000	217	6	1	0.714285714	0.7	1138	89,156.0		\$50	46021473	1	2	1	
	consistent	WTF	World Taekwondo Championships	448000	503249	68300	182	6	0	0.5174285714	0.9	36	63,037.0		\$0	7537271	1	2	1	
Table Tennis	consistent	ITP	Worldnetball	78000	30400	5400	211	6	0	0.5174285714	0.5	706	53,047.0		\$5	4820281	1	2	1	
	consistent	ITF	ITF World Tennis Championships	310000	108000	160	6	0	0.4285714286	0.8	1875	67,125.1		\$90	792385	1	2	1		
Volleyball	consistent	FIVB	FIVB Volleyball World Grand Champions Cup	3340000	270000	493500	218	6	0	0.4285714286	0.7	4061	21,176.1		\$30	6526342	1	2	1	
	consistent	WBF	World Wrestling Championships	76100	184000	12400	152	6	0	0.4285714286	0.9	1287	73,155.1		\$100	1351021	1	1	1	
Wrestling	consistent	UWW	World Wrestling Championships	899000	178200	18000	130	6	0	0.28174286	0.8	409	152.96		\$0	24185107	1	2	1	
	consistent	WR	World Rugby Cup	180000	160000	56000	130	6	0	0.28174286	0.7	820	6,027.0		\$90	2000510	1	2	1	
Baseball and Softball	2020 jan	World Skate	Rugby World Cup	8000	151000	7979	100	6	0	0.4285714286	0.7	261	77,122.9		\$100	11648430	1	2	1	
	2020 jan	IFSC	IFSC Climbing World Cup	363000	283000	42100	183	6	0	0.4285714286	0.7	680	75,24.0		\$40	2657045	1	2	1	
Skateboarding	2020 jan	ISA	Red Bull World Surf League Finals	51400	224000	20800	60	6	0	0.4285714286	0.6	244	63,035.0		\$400	933400	1	2	1	
	2028 jan	WBSC	World Baseball Classic	195000	157000	119100	130	6	0	0.4285714286	0.6	1278	58,13.7		\$350	1831383	1	2	1	
Cricket	2028 jan	ICC	ICC Men's Cricket World Cup	310000	310000	290000	108	6	0	0.4285714286	0.8	1448	88,122.0		\$120	1310821	1	2	1	
	2028 jan	IFAF	IFAF Flag Football World Championship	0	0	991	75	6	0	0.5174285714	0.7	775	75,26.0		\$40	13031029	1	2	1	
Squash	2028 jan	WSP	the PSA World Championship	14800	22000	17500	130	6	1	0.5174285714	0.8	430	84,18.0		\$30	1316555	1	2	1	
	2024 q4	WDSF	World Breaking Championships	27000	115253	6542	1	6	0	0.4185714286	0.7	62	107.87		\$140	7481465	1	2	1	
Baseball	2028 quit	IBAF	IBA World Baseball Championship	308000	390000	312000	130	6	0	0.4285714286	0.7	4547	68,033.0		\$100	10327261	1	2	1	
	2024 quit	WBFA	World Karate Federation Championships	740000	430000	48300	198	6	0.8	0.540	104,29.0		\$160	161723	1	2	0	1		
Football	highly unstable	UEFA	UEFA Champions League	10000000	28400000	12000000	2051	6	0	0.4285714286	0.1	41	10979,129.0		\$1000	63627948	1	2	1	
	highly unstable	Formula 1	Formula 1	178000	2,211,517	1000000	245	6	1	0.4185714286	0.0	325	62,42.6		\$15,000,000	1872480	0	2	0	
Basketball	highly unstable	WBA	World Boxing Championships	19000	80211	60000	115	6	0	0.4285714286	0.1	418	61,613.0		\$100	1342665	1	2	1	
	highly unstable	WFA	World Wrestling Championships	138000	421000	123000	140	6	0	0.4285714286	0.1	0.4285714286	95,0.0		\$1000	1787480	1	2	0	
Baseball	highly unstable	World Pool Federation	World Pool Championship	481	12	12	6	0	0.4285714286	0.1	0.4285714286	95,0.0		\$100	1342665	1	2	0		
	highly unstable	World Martial Arts Organization	World Martial Arts Championship	0	0	0	12	1	0	0.4285714286	0.1	0.4285714286	95,0.0		\$100	1342665	1	2	0	
Baseball	highly unstable	World Karate Federation	World Karate Championships	116000	130000	90511	120	6	0	0.8174285714	0.8	92	108.1		\$10	2051008	0	1	0	
	highly unstable	WBA	World Boxing Championships	19000	80211	60000	115	6	0	0.4285714286	0.1	418	61,613.0		\$100	1342665	1	2	1	
Baseball	highly unstable	International Judo Federation	World Judo Championships	850	0	0	90	6	0	0.28174286	0.8	2	1.00		\$0	23288	1	2	1	
	highly unstable	World Archery	World Archery Championships	610000	330000	81200	140	0	1	0.8174285714	0.6	919	68,153.1		\$0	6181811	1	2	1	
Baseball	highly unstable	World Taekwondo	World Taekwondo Championships	448000	503249	68300	182	6	0	0.5174285714	0.9	36	63,037.0		\$0	7537271	1	2	1	
	highly unstable	World Wrestling	World Wrestling Championships	76100	184000	12400	152	6	0	0.4285714286	0.9	1287	73,155.1		\$100	1351021	1	1	1	
Baseball	highly unstable	World Bridge Federation	World Bridge Championships	0	1899	0	100	6	0	0.4285714286	0.1	0.4285714286	95,0.0		\$10	1655935	1	2	1	
	highly unstable	International Ice Federation	World Ice Hockey Championships	510	79	0	12	6	0	0.28174286	0.6	0	4,810.0		\$10	174862	1	2	1	
Baseball	highly unstable	nan	Chester's Milk Cheese Rolling and Wake	0	0	0	0	0	0	0.8174285714	0.8	0	3.0		\$0	2001075	1	2	1	
	highly unstable	International Rugby Federation	World Rugby Cup	786	21883	2996	60	6	0	0.5174285714	0.1	520	10,1.0		\$0	109111	1	2	0	
Baseball	Not consistent	WCF	World Curling Championships	0	0	0	60	6	0	0.5174285714	0.6	66	284.0		\$0	56611	1	2	0	
	Not consistent	French Open	French Open	0	0	0	35	6	0	0.4185714286	0.3	46	1.00		\$0	1000033	1	2	1	
Baseball	Not consistent	FIP	French Open	103000	22000	17500	130	6	1	0.5174285714	0.8	430	84,18.0		\$30	1316555	1	2	1	
	Not consistent	British Open	British Open	2950	158	1214	18	5	0	0.4185714286	0.5	71	1.00		\$0	690878	1	2	1	
Baseball	Not consistent	TWIF	Tag of War World Championships	0	1224	0	30	6	0	0.8174285714	0.7	1	10,819.0		\$0	99900	1	2	1	
	Not consistent	UIM	UIM Formula 1 Powerboat World Championships	770	3796	704	6	0	0.4185714286	0.1	0.4185714286	95,0.0		\$117,000	16118	1	2	0		
Baseball	Considerations	WTF	World Taekwondo Championships	448000	503249	68300	182	6	0	0.5174285714	0.9	36	63,037.0		\$0	7537271	1	2	1	
	Considerations	WFA	World Wrestling Championships	76100	184000	12400	152	6	0	0.4285714286	0.9	1287	73,155.1		\$100	1351021	1	1	1	
Baseball	Considerations	IFMA	IFMA World Muaythai Championships	8380	15400	6918	128	6	0	0.4185714286	0.5	217	1,01.5		\$100	2137763	1	2	1	
	Considerations	IFJ	World Judo Championships	514000	130000	99700	208	6	0	0.4285714286	0.8	424	79,321.0		\$0	984143	1	2	1	
Baseball	Considerations	WCF	World Curling Championships	0	0	0	60	6	0	0.5174285714	0.6	66	284.0		\$0	56611	1	2	0	
	Considerations	WFA	World Wrestling Championships	76100	184000	12400	152	6	0	0.4285714286	0.9	1287	73,155.1		\$100	1351021	1	1	1	
Baseball	Considerations	WCF	World Curling Championships	0	0	0	60	6	0	0.5174285714	0.6	66	284.0		\$0	56611	1	2	0	
	Considerations	WFA	World Wrestling Championships	76100	184000	12400	152	6	0	0.4285714286	0.9	1287	73,155.1		\$100	1351021	1	1	1	
Baseball	Considerations	WCF	World Curling Championships	0	0	0	60	6	0	0.5174285714	0.6	66	284.0		\$0	56611	1	2	0	
	Considerations	WFA	World Wrestling Championships	76100	184000	12400	152	6	0	0.4285714286	0.9	1287	73,155.1		\$100	1351021	1	1	1	
Baseball	Considerations	WCF	World Curling Championships	0	0	0	60	6	0	0.5174285714	0.6	66	284.0		\$0	56611	1	2	0	
	Considerations	WFA	World Wrestling Championships	76100	184000	12400	152	6	0	0.4285714286	0.9	1287	73,155.1		\$100	1351021	1	1	1	
Baseball	Considerations	WCF	World Curling Championships	0	0	0	60	6	0	0.5174285714	0.6	66	284.0		\$0	56611	1	2	0	
	Considerations	WFA	World Wrestling Championships	76100	18															